

Real Time Processing of Musical AI Models

Alex Miale, Gautham Srinivasan, Avery Huang, and Prof. Victor Zappi

Background, Motivation, and Goals:

- AI is increasingly prominent in **music production**, with machine learning (ML) tools offering new creative outcomes, high-quality processing, and sped-up workflows
- Substantial **computational demands** combined with the need for **real-time processing** leads developers to create **limited** but still **resource-intensive** models, forcing musicians into **costly upgrades** or **limiting capabilities**
- The goal is to develop **novel ML solutions** that run AI music models on **hardware accelerators** commonly found on modern chipsets

Process and Methods:

1. Identify relevant literature with associated models in **Python** and **PyTorch**, an open-sourced deep learning library and framework
2. Run the models in dedicated **inference engines** for real-time processing
3. Embed models in **development frameworks** in **C++** language

Findings and Products:

4. CAMD Day 2026 demonstration of of distortion model on phone application using a guitar
- Performance panel demonstrates graphs of computational time

Conclusions and Next Steps:

- Continue to research ML audio models for different effects like **Phasers** and **Flangers**
- Continue to embed current models on to different hardware accelerators like **GPUs** and **DSPs**

Acknowledgements:

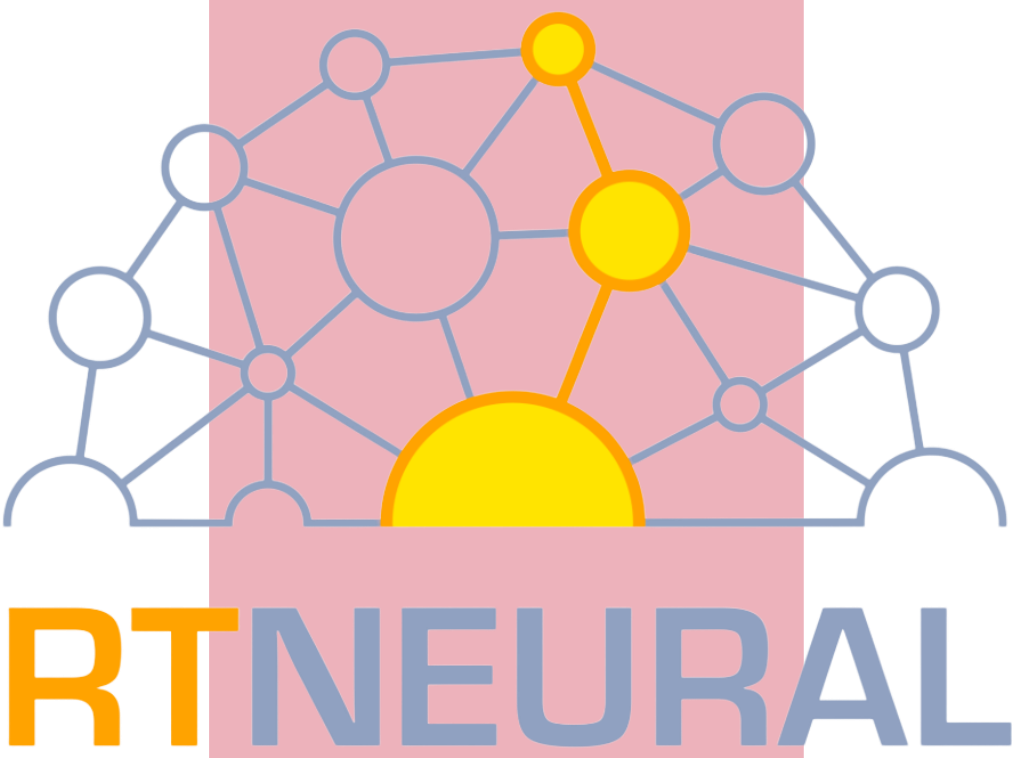
- This research is funded by the PEAK Summit Award from Undergraduate Research and Fellowships at Northeastern University

1

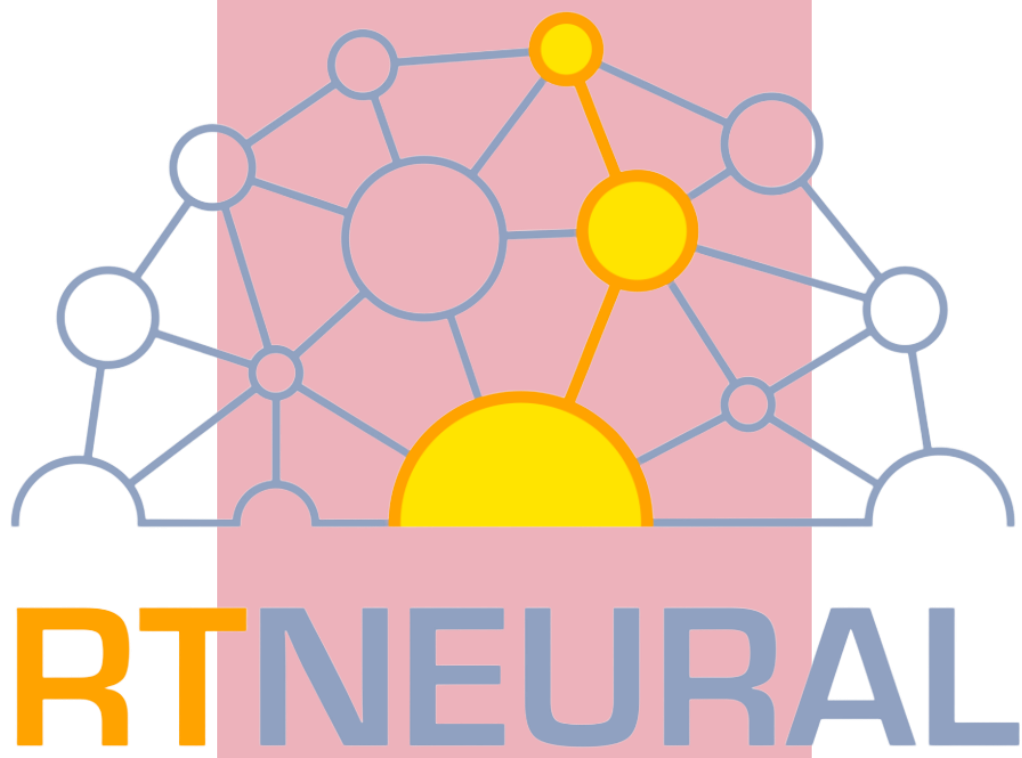


We designed a unified workflow to turn **Musical AI Models** from research artifacts into usable applications **across devices**

2



3

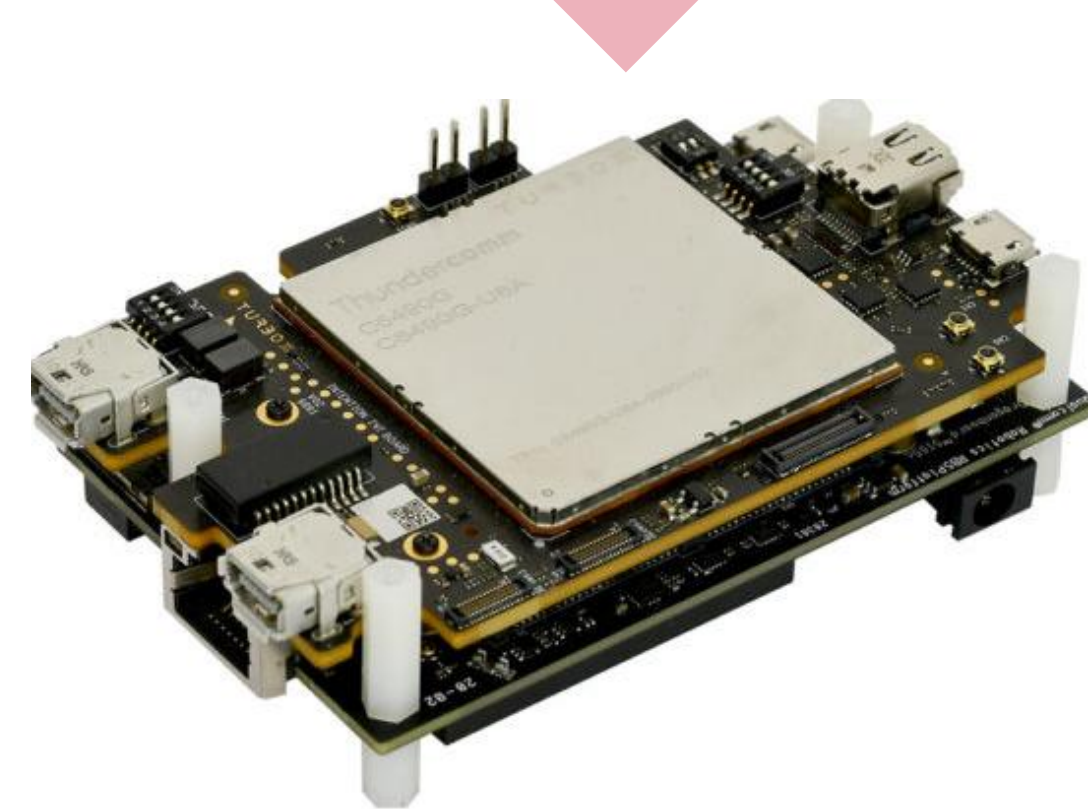


LDSP



Qualcomm
QNN

AudioReach



4

Phone Guitar Pedal



Benchmarking

Models

Model	Task	S - Rate	Layers	Params
NAM	Amp Modelling	48 kHz	Conv 1D	13,801
MS-Wavehax	Vocoding	48 kHz	Conv 2D	~500k
BRAVE	Timbre Transfer	44.1 kHz	VAE	~6M

Protocol

5 runs for 10 seconds each
on RB3 Gen2 board;
Measured inference time
Mean + Std

Results

